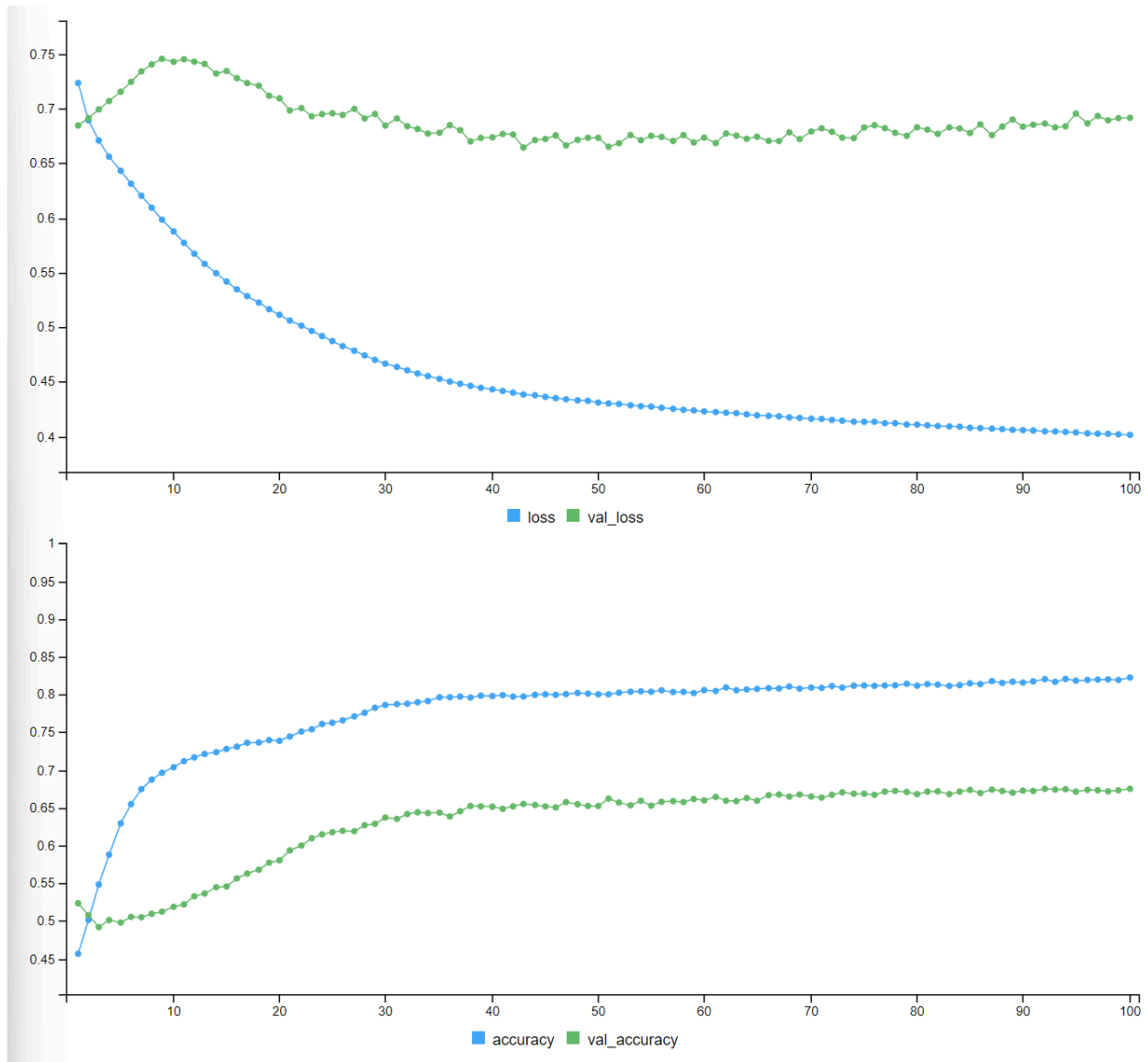


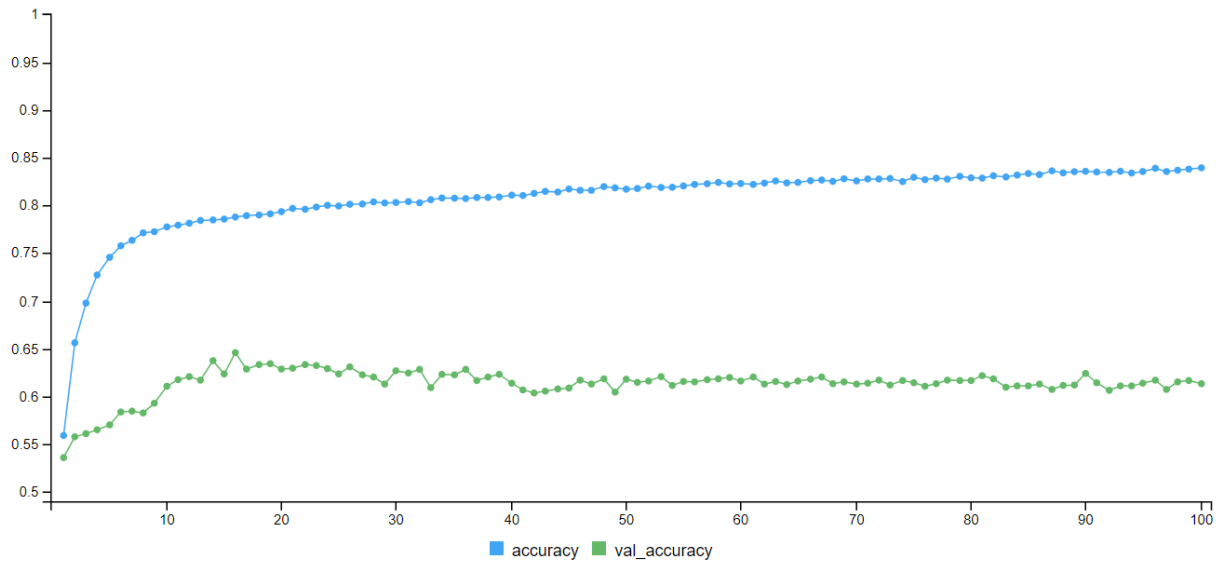
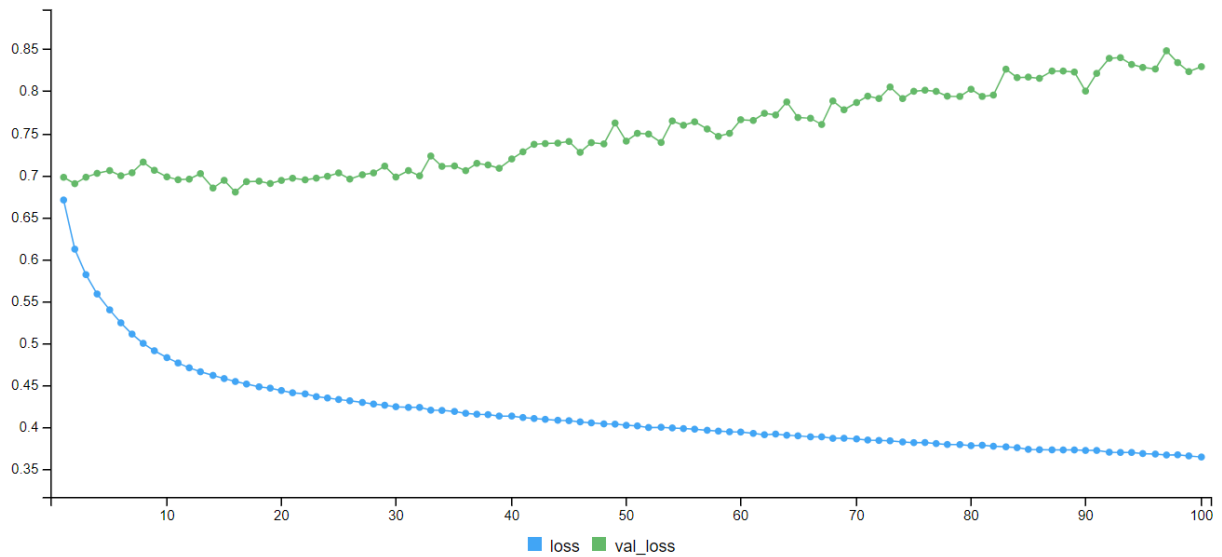
Module 04: Lab 04

Exercises

1) Copy and paste the loss and accuracy curves obtained from running the code above (note, that the curves will be slightly different than those shown in this lab).



2) Change the hidden layers to have 50 units and 25 units, respectively, and re-run the code. Copy and paste the new loss and accuracy curves.



3) Compare the curves from 1) and 2) and discuss which architecture (i.e., number of nodes in the hidden layers) results in better performance.

Training Accuracy:

- **Original Model:** The training accuracy gradually increases and stabilizes around 81.6%.
- **Modified Model:** The training accuracy improves quickly and stabilizes around 82.3%.

Validation Accuracy:

- **Original Model:** The validation accuracy goes up to 81.8%.
- **Modified Model:** The validation accuracy goes up to 82.4%.

Training Loss:

- **Original Model:** The training loss decreases and stabilizes around 0.38.
- **Modified Model:** The training loss decreases and stabilizes around 0.37.

Validation Loss:

- **Original Model:** The validation loss stabilizes around 0.38.
- **Modified Model:** The validation loss stabilizes around 0.37.

The modified model with more units in the hidden layers (50 and 25 units) shows improvement in both training and validation accuracy compared to the original model (20 and 10 units) as well as for the loss curves where our modified model performs slightly better, with lower final training and validation losses.

4) Calculate the accuracy on the test set for the models in 1) and 2). Which accuracy is better?

Original
loss accuracy
0.5834309 0.7379532

Modified
loss accuracy
0.6003162 0.7384121

The modified test set has better accuracy.